University of Toronto

# Inference-Time Adaptive Execution for Transformer Action Chunking

CSC415 Course Project Final Presentation

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CSC415: Introduction to Reinforcement Learning

April 2026

# Contents


- Background
- Motivation
- Research gap & project aim
- Methodology
- Experimental setup
- Results & discussion
- Limitations and future work

# Background

- **Action Chunking with Transformers** by Zhao et al. (2023) is a method that combines transformers (CVAE) and action chunking for great long-horizon performance and stability.
- **RoboMimic** is a suite of exercises centered around simulating a robotic manipulator arm around various tasks

# Motivation

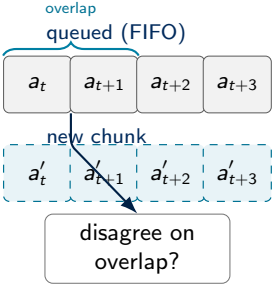
- Action Chunking provides temporal abstraction by predicting $k$ future actions at once instead of predicting single actions, which reduces the length of the horizon by $k$ times.
- Transformer action chunking improves long-horizon control, but fixed chunk length can reduce reactivity near contact events.
- Real-time execution can show chunk-boundary inconsistency/jitter under latency and uncertainty.
- We focus on **inference-time execution policy**: when should a chunk be preempted and replanned?
- Goal: improve success/reactivity tradeoff without changing simulator dynamics or requiring extra labels.

# Research gap & project aim

- Prior chunking methods usually fix execution horizon or use smoothing, but do not deeply study adaptive commits in our low-dim RoboMimic setup.
- We propose **adaptive commit-length** execution using disagreement / delta / uncertainty rules.
- Core hypothesis: adaptive execution can match or improve rollout quality while reducing unnecessary replans.

# Methodology

- ACT policy predicts a chunk $(a_t, \ldots, a_{t+k_{\max}-1})$.
- Executor holds a **FIFO** action queue; commit length $m \in [k_{\min}, k_{\max}]$.
- **Overlap disagreement:** compare the still-queued tail with the overlapping prefix of the *new* chunk; if they disagree enough, **preempt** and replan.
- Ablations: action-change threshold, stochastic uncertainty threshold.
- Temporal ensembling included as a comparator.



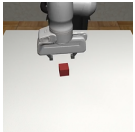
# Experimental setup

- Tasks: RoboMimic low-dim Lift, Can, Square.
- ACT-Strong config shared across tasks; matched eval protocol.
- Baselines: BC-MLP, kNN-BC (VINN-style), BeT-style discrete BC.
- Metrics: success, return, policy queries, replans/episode, commit statistics.
- Rigor: ablations + 3-seed runs.

# ACT-Strong results (cross-task)

| Task / mode | Success | Return |
|---|---|---|
| Lift dynamic | 1.00 | 60.00 |
| Lift temporal ensemble | 1.00 | 76.10 |
| Can dynamic (z0) | 0.50 | 144.90 |
| Can temporal ensemble (z0) | 1.00 | 289.15 |
| Square dynamic (z0) | 0.15 | 42.05 |
| Square receding (z0) | 0.20 | 41.70 |

- Lift strongly supports adaptive execution.
- Can is mixed and setting-sensitive.
- Square is challenging.



Lift

Can

Square

# Baseline summary (receding-horizon eval)

| Task / baseline | Success | Return |
|---|---|---|
| Lift BC / kNN / BeT | 0.65 / 0.75 / 0.75 | 70.7 / 44.4 / 127.3 |
| Can BC / kNN / BeT | 0.00 / 0.30 / 0.30 | 0.0 / 58.65 / 33.75 |
| Square BC / kNN / BeT | 0.05 / 0.35 / 0.05 | 0.3 / 43.0 / 5.6 |

- Baselines are competitive on Lift, weaker on Can/Square.
- Supports increased task difficulty from Lift to Can/Square in this setup.

# Limitations and future work

- Limited time, storage
- Hypothesis outcome: **partially supported** (strong Lift, mixed Can, weak Square).
- Dynamic rules likely need task-specific threshold calibration.
- Future work: threshold sweeps, uncertainty calibration, and stronger representation learning for Square.
- Add significance tests / confidence intervals across more seeds where compute allows.

# References I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# Thank you for your attention

**Questions?**